

Artificial Intelligence and Machine Learning

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

5133

Credits

4

Periods

5/week

Course Details

Title: Artificial Intelligence and Machine Learning

Department: Common

Revision: REV_Artificial Intelligence and Machine Learning

Pattern: Theory / Practical

Complies with official SITTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 5133 — Artificial Intelligence and Machine Learning**. Use the official SITTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ു ഐഐഐഐഐഐ ഐഐഐഐ ഐഐഐഐ English technical terms- ഐഐഐഐഐഐ ഐഐഐഐഐഐ. ുഐ ഐഐഐഐഐ ഐഐഐഐഐഐ ഐഐഐഐഐഐ, formula, diagram, units, definitions ഐഐഐഐ ഐഐഐഐഐഐഐഐ ഐഐഐഐഐഐഐഐ.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Artificial Intelligence and Machine Learning.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

Core Principles

Master foundational terminology and core definitions of Artificial Intelligence and Machine Learning.

Detailed analysis of core principles and theoretical foundations.

MODULE 2

Applied Methods

Apply standard calculation techniques and operational procedures.

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഐക്യം നമ്മുടെ ശക്തിയാണ്. കൂടെ ചേർന്ന് പ്രശ്നങ്ങൾ പരിഹരിക്കാം.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ഏകീകരണത്തിനും പ്രവർത്തനം മെച്ചപ്പെടുത്തുന്നതിനും സങ്കീർണ്ണ സംവിധാനങ്ങൾ വിലയിരുത്തുകയും പരിഹരിക്കുകയും ചെയ്യുന്നതിനുള്ള ഉപാധികളും സാങ്കേതികവിദ്യകളും പഠിക്കുകയും ചെയ്യുക.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: വിവിധ സാങ്കേതികവിദ്യകളെ സംയോജിപ്പിച്ച് പരിഹാരം തയ്യാറാക്കുന്നതിനും പ്രവർത്തനം മെച്ചപ്പെടുത്തുന്നതിനും സാങ്കേതികവിദ്യകളെ പ്രദർശിപ്പിക്കുന്നതിനും പരിപാടികൾ പഠിക്കുകയും ചെയ്യുക.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Artificial Intelligence and Machine Learning.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.